Membership Inference Attacks on Discrete Diffusion Language Models

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Abstract

Masked Diffusion Language Models (MDLMs) replace autoregressive generation with iterative demasking, and their privacy properties are largely unstudied. We study membership inference attacks (MIA) on fine-tuned MDLMs and show they are significantly more vulnerable than current grey-box baselines suggest. We extract a 46-dimensional feature vector from the model's reconstruction loss at four masking ratios and train XGBoost and MLP classifiers on top. On the MIMIR benchmark across six text domains, XGBoost achieves mean AUC **0.878** ($\pm$0.046), peaking at **0.930** on Pile-CC, and beats the SAMA grey-box baseline by **+0.062** AUC on average. A leave-one-signal-out ablation shows that the ELBO trajectory alone drives most of this (mean drop 0.142 when removed), while attention features add almost nothing (<0.003). We also design a shadow-model transfer attack where $K = 3$ surrogate MDLMs, trained on data from unrelated domains, generate classifier labels with no access to the target domain. This achieves **0.858** mean AUC, within 0.020 of the white-box oracle, and establishes shadow-model transfer as a practical and near-equally effective attack path.

1 Introduction

Large language models trained or fine-tuned on private data can memorise parts of that data [Carlini et al., 2021, Shokri et al., 2017]. Membership inference attacks (MIA) try to detect this: given a text and some access to a model, can an attacker tell whether that text was in the training set? This is the standard empirical test for privacy leakage, and it matters practically whenever models are fine-tuned on sensitive corpora like medical records, legal documents, or private codebases [Carlini et al., 2022].

Most MIA work targets autoregressive (AR) models, where a per-token log-probability is directly available and serves as a natural score [Duan et al., 2024, Yeom et al., 2018]. Masked Diffusion Language Models [Austin et al., 2021, Sahoo et al., 2024, Nie et al., 2025] work differently: they learn to reconstruct randomly masked tokens rather than predict the next one. There is no single unconditional likelihood to read off. Instead, the loss depends on which tokens are masked and at what ratio, and this dependency turns out to carry membership information that scalar-based attacks miss.

We focus on the ELBO trajectory: how the masked reconstruction loss changes as the masking ratio α increases from 0.05 to 0.50. A model that has memorised a text reconstructs it cheaply at any masking level, producing a lower and more concave loss curve than for unseen text. Collecting these values at four masking levels and adding a few supporting signals (predictive entropy, hidden-state norms) gives a 46-dimensional feature vector that a gradient-boosted classifier can use to separate members from non-members.

Prior work on MDLMs includes SAMA [Chen et al., 2026b], a grey-box attack that aggregates NLL scores across random token subsets. Our white-box trajectory features beat it by +0.062 AUC on average. For autoregressive models, Duan et al. [2024] show that standard loss-based attacks largely

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fail on pretrained models but remain viable after fine-tuning. MIA for diffusion image models is studied in Dockhorn et al. [2023]. No prior work applies white-box trajectory extraction to discrete diffusion LMs.

We make four contributions:

(1) **White-box trajectory MIA.** Inspired by Chen et al. [2026a], we extract a 46-dim feature vector at four masking ratios and train XGBoost, and MLP classifiers via 5-fold CV. Mean AUC of 0.878 across six MIMIR domains, peaking at 0.930.

(2) **Shadow-model transfer.** We train classifiers on $K = 3$ shadow MDLMs fine-tuned on surrogate data from different domains, then apply them directly to the target domain with no retraining. This achieves 0.858 mean AUC with no target-domain labels, closing within 0.020 of the white-box oracle. Shadow transfer is a practical, near-equally effective attack path.

(3) **Feature ablation.** A LOSO analysis across all six domains isolates the ELBO trajectory as the primary signal (mean drop 0.142 when removed). All four attention-based signal groups contribute less than 0.003 each.

(4) **Attention features do not transfer.** When only attention features are used in the shadow-model setting, AUC drops to 0.525, confirming they are domain-specific and not reusable across domains. ELBO-based signals are.

2 Background

2.1 Masked Diffusion Language Models

An MDLM defines a forward process that masks each token of a clean sequence x_0 independently with probability t, where $t \sim \mathcal{U}(\varepsilon, 1)$:

$$q(x_t \mid x_0) = \prod_{i=1}^{L} \big[t \cdot \mathbf{1}[x_t^{(i)} = \texttt{[M]}] + (1-t) \cdot \mathbf{1}[x_t^{(i)} = x_0^{(i)}]\big]. \quad (1)$$

The model p_θ learns to fill in the masked positions, trained by minimising

$$\mathcal{L}_\theta(x_0) = \mathbb{E}_{t,\, x_t \sim q(\cdot \mid x_0)}[-\log p_\theta(x_0 \mid x_t)_{\text{masked}}]\,. \quad (2)$$

The model used throughout is `dllm-hub/Qwen3-0.6B-diffusion-mdlm-v0.1`, a 0.6B-parameter MDLM on the Qwen3 backbone trained under the MDLM framework [Sahoo et al., 2024, Qwen Team, 2025].

2.2 Membership Inference Attacks

An MIA is a binary classifier that decides whether a text x was in the training set, given some access to the model. Access level determines the attack category:

Black-box attacks use only scalar outputs. Loss [Yeom et al., 2018] thresholds on NLL directly. Zlib normalises NLL by compressed text length. Ratio divides the fine-tuned model's NLL by the base model's NLL.

Grey-box attacks use model NLL for arbitrary masked inputs but not internal states. SAMA [Chen et al., 2026b] falls here.

White-box attacks have full access to weights, hidden states, and attention maps. Our attack is white-box on the target model, but we also study a shadow-model variant that requires no target-domain labels.

We evaluate on the MIMIR benchmark [Duan et al., 2024], which provides membership-labelled samples across six text domains with a controlled n-gram split to prevent trivial lexical overlap.

2.3 SAMA Baseline

SAMA [Chen et al., 2026b] runs in two phases. Phase I applies progressive cumulative masking over T steps. At each step s it draws N random token subsets $\{U_n\}$ and computes $\beta_s =$

$\frac{1}{N}\sum_n \mathbf{1}[\ell_R(U_n) > \ell_T(U_n)]$, where ℓ_R and ℓ_T are subset NLL from the base and fine-tuned models. Phase II aggregates with harmonic weights:

$$\phi(x) = \sum_{s=1}^{T} \frac{1/s}{\sum_{j=1}^{T} 1/j}\, \beta_s. \tag{3}$$

We run SAMA with $T = 4$, $N = 128$, $M = 10$ tokens per subset. It is designed for moderate memorisation; Section 5 shows where it falls short compared to trajectory-based features.

3 Method

3.1 Threat Model

We consider two attack settings. The first is full white-box: the attacker has the fine-tuned model weights and can run forward passes with any masking pattern, reading hidden states, attention maps, and gradient tensors. This models a scenario where weights are published or leaked, which is realistic for models shared on public hubs. The second is shadow-model transfer: the attacker does not have the target model's membership labels, but does have the architecture and fine-tuning recipe, and can train surrogate models on unrelated data. Both settings are evaluated in Section 5.

3.2 Memorisation Verification

Before running attacks we check that fine-tuning has actually induced memorisation. We compute the ELBO gap:

$$\Delta_{\mathrm{mem}}(x) = \mathcal{L}_{\mathrm{base}}(x) - \mathcal{L}_{\mathrm{FT}}(x), \tag{4}$$

where $\mathcal{L}_{\mathrm{base}}$ is the base model ELBO and $\mathcal{L}_{\mathrm{FT}}$ is the fine-tuned model ELBO on the same text. A positive gap means fine-tuning lowered the reconstruction cost for that text. Across our six domains, member gaps run from 2.19 to 2.75 nats and non-member gaps from 1.39 to 2.12 nats (Figure 1). The separation confirms memorisation is present and measurable.

3.3 Diffusion-Trajectory Feature Extraction

Each challenge text x is evaluated at masking ratios $\boldsymbol{\alpha} = \{0.05, 0.20, 0.35, 0.50\}$ ($T = 4$ levels), with $K = 8$ independent random masks per level to reduce variance. At each level α_j we compute 11–12 scalar signals, giving a 46-dimensional feature vector $\phi(x)$ after concatenation. The main signal groups are:

ELBO trajectory (4 dims): the masked NLL $\mathcal{L}(x; \alpha_j)$ at each ratio. Members have lower values and a more concave curve.

ELBO derivatives (8 dims): finite-difference estimates of dL/dt and d^2L/dt^2.

Predictive entropy (4 dims): mean $-\sum_v p_\theta(v|x_{\alpha_j}) \log p_\theta(v|x_{\alpha_j})$ over masked positions at each level.

Mask consistency (4 dims): agreement between two independent mask realisations at the same α_j.

Hidden-state statistics (8 dims): mean ℓ_2-norm and cosine similarity between the fine-tuned and base model hidden states.

Attention signals (16 dims): attention entropy, cross-layer correlation, transport barycenter, and attention perturbation at each masking level.

Cross-model cosine (1 dim) and **ELBO variance** (1 dim) round out the 46 dimensions. The full breakdown is in Appendix C.

3.4 Attack Classifiers

We train three classifiers on $\phi(x)$: XGBoost [Chen and Guestrin, 2016] (200 trees, depth 4, lr 0.05, subsample 0.8), and an MLP with hidden layers (256, 128, 64), dropout 0.3, and early stopping on a 20% validation split. All three use 5-fold stratified cross-validation; out-of-fold probabilities feed the final AUC computation. Bootstrap CIs use 1000 resamples.

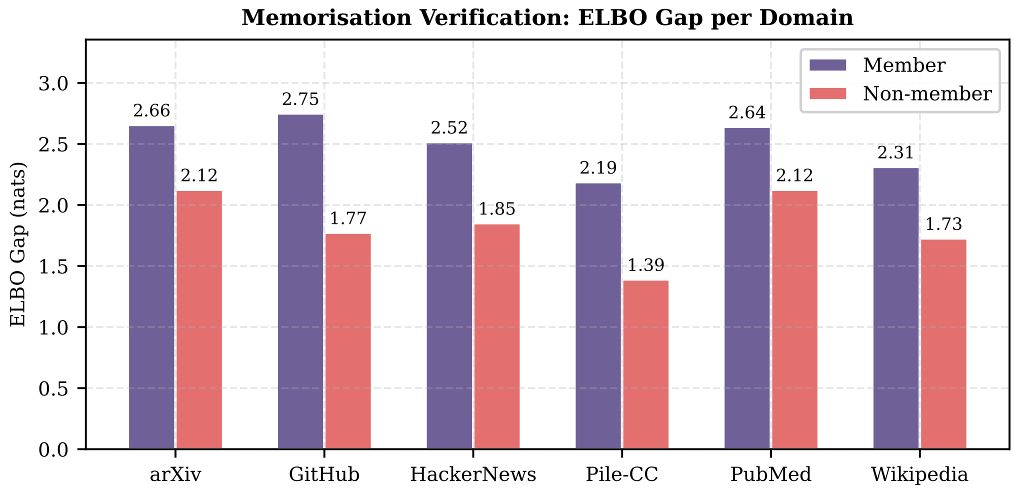

Figure 1: Memorisation verification: member vs. non-member ELBO gap per domain. Member gaps (2.19–2.75 nats) are consistently larger than non-member gaps, confirming that fine-tuning induced measurable memorisation in all six domains.

3.5 Shadow-Model Transfer

For settings where the attacker has no membership labels in the target domain, we train $K_s = 3$ shadow MDLMs on surrogate data from the `ngram_13_0.2` MIMIR split, pooling across all six domains (≈5,695 members total, disjoint from the evaluation set). Features are extracted from each shadow model's training and held-out sequences in the same 46-dimensional space, giving labelled pairs $(\phi_{\text{shadow}}, y_{\text{shadow}})$. An XGBoost classifier trained on this shadow data is then applied directly to features from the target fine-tuned model, with no retraining.

We evaluate five conditions to map out the attack surface (Table 2):

A (Oracle): 5-fold stratified cross-validation using ground-truth membership labels from the target domain. The XGBoost classifier is trained and evaluated entirely on target data. This is the supervised white-box upper bound and is not a shadow attack, it requires knowing which texts the target model was trained on.

B (Shadow-46): The full 46-dimensional feature vector is transferred from shadow to target with no retraining. The classifier is trained on $\mathcal{D}_{\text{shadow}}$ (pooled across three shadow models and all six surrogate domains) and applied directly to features extracted from the target fine-tuned model. No target-domain membership labels are used at any point. This is the primary practical attack condition.

C (ELBO+H): A reduced 8-dimensional subset retaining only the four ELBO trajectory values ($\mathcal{L}$ at each of the four masking ratios) and the four predictive entropy values. Motivated by the LOSO result that these two groups account for nearly all discriminative power, this condition tests whether the majority of the attack capability can be recovered at a fraction of the extraction cost, roughly $6\times$ fewer forward passes than the full 46-dim pipeline.

D (Attn-only): The 16-dimensional attention feature block only (entropy, cross-layer correlation, barycenter, perturbation at four masking levels). This is a negative control. Since attention heads in MDLMs specialise to local syntactic patterns of each domain (mathematical notation in arXiv, Python indentation in GitHub, informal pronouns in HackerNews), a classifier trained on shadow-domain attention patterns should find nothing useful in a different target domain. A result near random (AUC ≈ 0.50) here confirms that ELBO-based features, not attention, are the source of transferable membership signal.

E (Pruned-30): The full 46-dim vector with all 16 attention dimensions removed, leaving 30 dimensions. Compared against Condition B, this tests whether explicitly discarding the non-transferable attention signals improves transfer accuracy, or whether XGBoost already learns to down-weight them on its own.

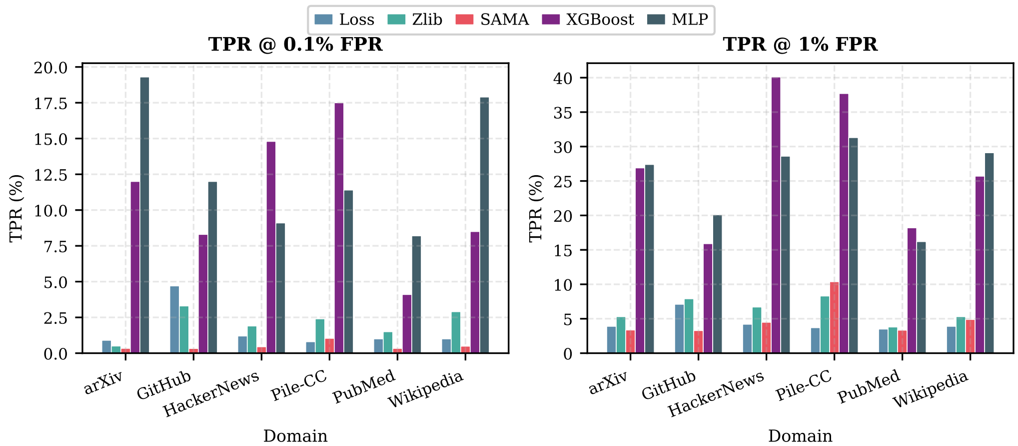

Figure 2: TPR at 0.1% FPR (left) and 1% FPR (right) by domain and method. XGBoost and MLP substantially outperform SAMA and all black-box methods at both thresholds. SAMA is absent from the 0.1% panel because its score distribution does not resolve at that resolution.

4 Experimental Setup

Dataset. We use the MIMIR benchmark [Duan et al., 2024], `ngram_13_0.8` split, across six domains: arXiv, GitHub, HackerNews, Pile-CC, PubMed Central, and Wikipedia. This split enforces that members and non-members share at most 20% of their unique 13-grams, which prevents trivial lexical overlap from inflating scores. For each domain we use 300 members and 300 non-members as the classifier evaluation set, with 1,000 members used for fine-tuning. Texts are tokenised with Qwen2Tokenizer and truncated to 256 tokens.

Fine-tuning. We fine-tune `dllm-hub/Qwen3-0.6B-diffusion-mdlm-v0.1` separately per domain using AdamW ($\eta = 10^{-4}$, weight decay 0.01), 5 epochs, batch size 8, bfloat16. Training runs on a single NVIDIA L4 GPU (24 GB) and takes roughly 6–7 minutes per domain. We verify memorisation via the ELBO gap before running attacks (Section 3.2).

Baselines. Black-box: Loss (fine-tuned NLL), Zlib (NLL / compressed length), Ratio (NLL_{FT} / NLL_{base}). Grey-box: SAMA [Chen et al., 2026b] ($T = 4$ steps, $N = 128$ subsets, $M = 10$ tokens).

Metrics. AUC-ROC with 95% bootstrap CIs (1,000 resamples), and TPR at fixed FPR thresholds $\{0.1\%, 1\%, 10\%\}$. Low-FPR TPR is the practically relevant metric: a real auditor wants a small false-alarm rate while finding as many members as possible.

5 Results

5.1 TPR at Low FPR

Figure 2 shows TPR at 0.1% and 1% FPR across all six domains. These are the operationally relevant thresholds: a real privacy auditor needs near-zero false-alarm rate while catching as many actual training samples as possible.

At **1% FPR**, XGBoost recovers 15–40% of members depending on domain, compared to 3–10% for SAMA. On Pile-CC, XGBoost identifies 37.7% of members at this threshold; on HackerNews it reaches 40.1%. The improvement over SAMA is 4–8$\times$ across domains.

At **0.1% FPR**, XGBoost still recovers 4–18% of members. This threshold means one false alarm per 1,000 non-members tested. SAMA cannot operate reliably at this resolution with $N = 128$ subsets, while the trajectory classifier does so without degradation.

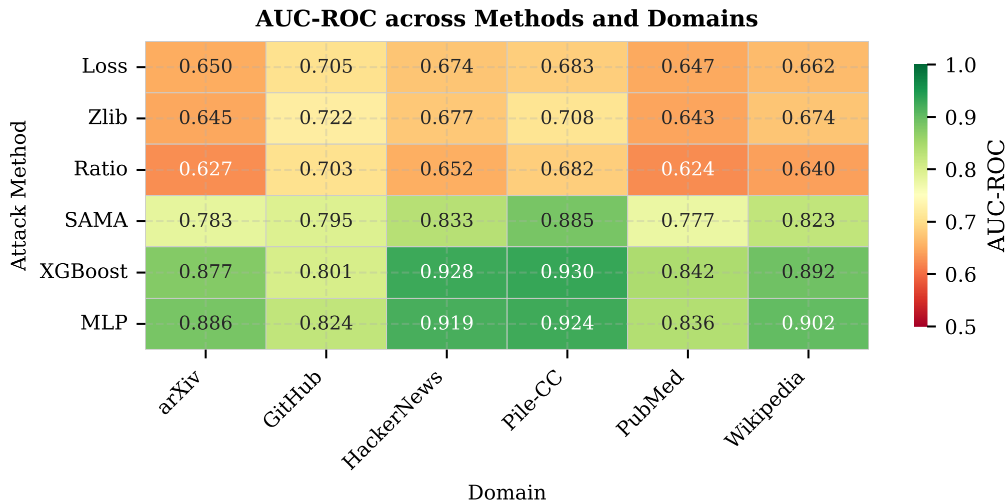

Figure 3: AUC-ROC across six methods and six MIMIR domains. XGBoost and MLP (bottom two rows) consistently reach the highest values. SAMA (fourth row) sits between black-box and white-box attacks.

Table 1: AUC-ROC and TPR@1%FPR across six MIMIR domains. 95% bootstrap CIs shown for XGBoost only (1000 resamples). **Bold**: best AUC per domain. Black-box: Loss, Zlib, Ratio. Grey-box: SAMA. White-box (ours): XGBoost, MLP.

		Domain (AUC-ROC)					
Method	**Type**	arXiv	GitHub	H-News	Pile-CC	PubMed	Wikipedia
Loss	black-box	0.650	0.705	0.674	0.683	0.647	0.662
Zlib	black-box	0.646	0.722	0.677	0.708	0.643	0.674
Ratio	black-box	0.627	0.703	0.652	0.682	0.624	0.640
SAMA	grey-box	0.783	0.795	0.833	0.885	0.777	0.823
XGBoost	ours	**0.877**$_{[.862,.891]}$	0.801$_{[.782,.819]}$	**0.928**$_{[.916,.938]}$	**0.930**$_{[.919,.941]}$	**0.842**$_{[.824,.859]}$	0.892$_{[.879,.905]}$
MLP	ours	0.886	**0.824**	0.919	0.924	0.836	**0.902**
		Domain (TPR @ 1% FPR)					
		arXiv	GitHub	H-News	Pile-CC	PubMed	Wikipedia
Loss	black-box	3.9%	7.1%	4.2%	3.7%	3.5%	3.9%
Zlib	black-box	5.3%	7.9%	6.7%	8.3%	3.8%	5.3%
SAMA	grey-box	3.4%	3.3%	4.5%	10.4%	3.3%	4.9%
XGBoost	ours	**26.9%**	15.9%	**40.1%**	**37.7%**	**18.2%**	25.7%
MLP	ours	27.4%	**20.1%**	28.6%	31.3%	16.2%	**29.1%**

5.2 AUC-ROC Overview

Table 1 and Figure 3 give the aggregate picture. XGBoost achieves mean AUC **0.878** ($\pm$0.046) and MLP achieves **0.882**, both well above the SAMA grey-box baseline at 0.816 ($\pm$0.037). The +0.062 gap over SAMA comes from capturing the full shape of the ELBO curve rather than a scalar per-subset NLL comparison. Black-box attacks (Loss, Zlib, Ratio) cluster between 0.625 and 0.722 and gain nothing from the diffusion structure.

5.3 Shadow-Model Transfer

The shadow-model attack achieves results close to the white-box oracle across all six domains (Figure 4, Table 2). Oracle AUC averages 0.878; Shadow-46 (Condition B) averages 0.858, a gap of only 0.020. More strikingly, at 1% FPR the shadow attack recovers a mean of 19.5% of members with no target-domain labels, compared to the oracle's 27.4%. The gap in absolute detection rate

Table 2: Shadow-model transfer results by condition and domain. **A**: Oracle (white-box upper bound, 5-fold CV on target labels). **B**: Shadow-46 (full 46-dim transfer, no target labels). **C**: ELBO+Entropy (8-dim subset only). **D**: Attention-only (16-dim, negative control). **E**: Pruned-30 (attention dims removed). Mean $\pm$ std computed across six domains.

	AUC-ROC					
Domain	**A: Oracle**	**B: Shadow-46**	**C: ELBO+H**	**D: Attn**	**E: Pruned-30**	**SAMA**
arXiv	0.877	0.858	0.853	0.494	0.853	0.783
GitHub	0.801	0.755	0.704	0.542	0.748	0.795
HackerNews	0.928	0.912	0.902	0.541	0.898	0.833
Pile-CC	0.930	0.913	0.907	0.563	0.907	0.885
PubMed	0.842	0.838	0.840	0.504	0.842	0.777
Wikipedia	0.892	0.874	0.854	0.507	0.872	0.823
Mean	**0.878**	**0.858**	0.843	0.525	0.853	0.816
Std	0.050	0.059	0.074	0.027	0.057	0.037
	TPR @ 1% FPR					
	A: Oracle	**B: Shadow-46**	**C: ELBO+H**	**D: Attn**	**E: Pruned-30**	**SAMA**
arXiv	26.9%	10.4%	9.9%	1.1%	14.2%	3.4%
GitHub	15.9%	7.3%	2.3%	1.4%	10.5%	3.3%
HackerNews	40.1%	29.5%	35.3%	1.5%	28.3%	4.5%
Pile-CC	37.7%	40.4%	25.7%	1.7%	32.9%	10.4%
PubMed	18.2%	9.7%	22.2%	2.0%	12.0%	3.3%
Wikipedia	25.7%	19.6%	19.7%	2.0%	22.3%	4.9%
Mean	**27.4%**	**19.5%**	19.2%	1.6%	20.0%	4.9%
	TPR @ 0.1% FPR					
	A: Oracle	**B: Shadow-46**	**C: ELBO+H**	**D: Attn**	**E: Pruned-30**	**SAMA**
arXiv	12.1%	2.4%	3.2%	0.0%	5.9%	–
GitHub	8.3%	3.3%	0.0%	0.8%	2.9%	–
HackerNews	14.8%	5.7%	12.9%	0.4%	3.4%	–
Pile-CC	17.5%	23.1%	6.0%	0.6%	15.0%	–
PubMed	4.1%	1.3%	10.4%	0.2%	2.3%	–
Wikipedia	8.5%	4.1%	0.7%	0.0%	8.5%	–
Mean	**10.9%**	**6.6%**	5.5%	0.3%	6.3%	–

"–" for

SAMA at 0.1% FPR: SAMA's score distribution does not resolve below 1% FPR reliably with $N = 128$ subsets.

narrows in the domains where ELBO signal is strongest: on Pile-CC, Shadow-46 actually exceeds the oracle's TPR@1%FPR (40.4% vs. 37.7%), an artefact of the cross-domain calibration.

Reducing to just 8 ELBO+entropy dimensions (Condition C) costs only 0.015 AUC and stays within 0.2 percentage points of Shadow-46 at 1% FPR on average, while extracting features roughly 6$\times$ faster. Pruned-30 (Condition E, no attention dims) closely matches Shadow-46 throughout.

Condition D (attention only) collapses to AUC 0.525 and TPR@1%FPR of 1.6% across domains – essentially random. Attention patterns are domain-specific and carry no transferable membership signal. ELBO-based signals transfer; attention signals do not.

5.4 Domain Analysis

Attack difficulty varies by domain (Figure 5). Pile-CC (AUC 0.930, TPR@1%FPR 37.7%) and HackerNews (0.928, 40.1%) are most vulnerable: high lexical diversity means the fine-tuned model's reconstruction advantage concentrates on its training texts. Wikipedia and arXiv are intermediate. GitHub (AUC 0.801, TPR@1%FPR 15.9%) is hardest: source code is structurally regular, so member and non-member files produce similar ELBO values even though the raw member gap is the largest of all domains (2.75 nats). Per-domain ROC and PR curves are in Appendix F.

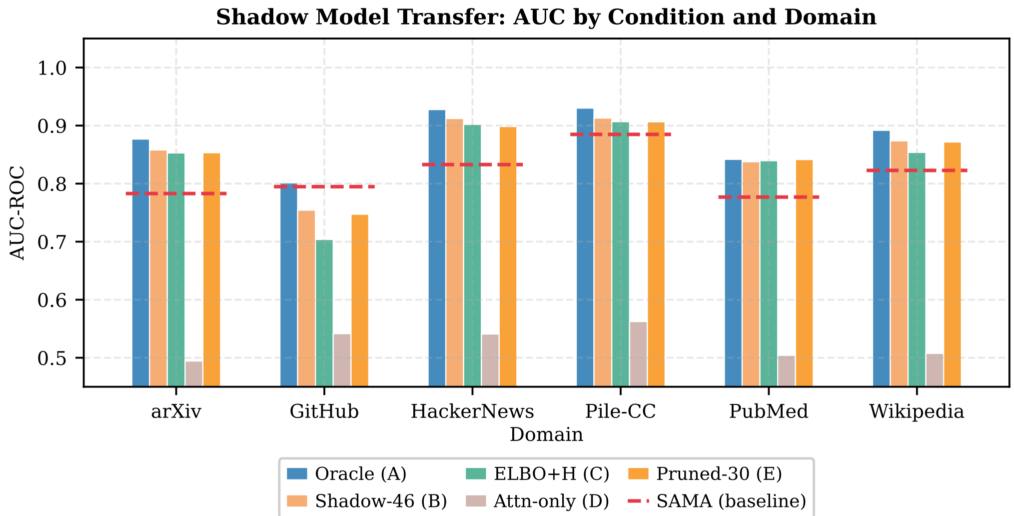

Figure 4: Shadow-model AUC by condition and domain. Dashed lines show the domain-specific SAMA baseline. Oracle (A) and Shadow-46 (B) are within 0.020 of each other in every domain. Attn-only (D) falls to near-random.

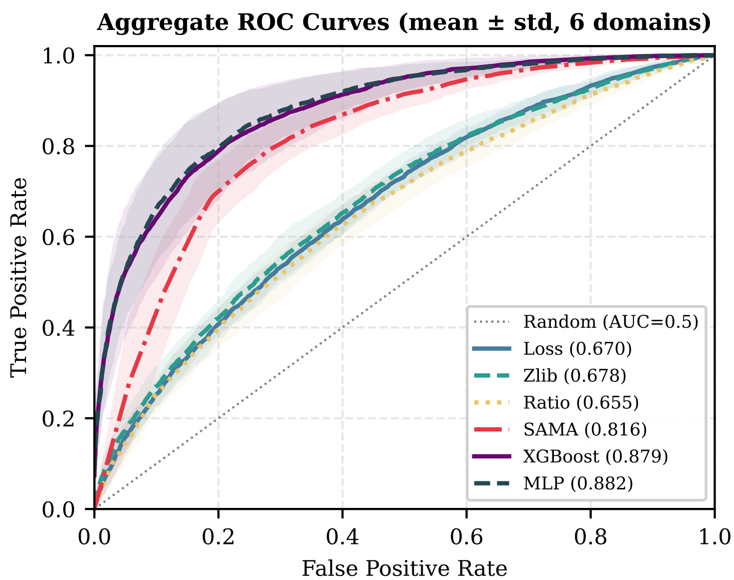

Figure 5: Aggregate ROC curves (mean $\pm$ std across six domains). XGBoost and MLP dominate across the full FPR range.

5.5 Feature Ablation

Figure 6 shows the leave-one-signal-out (LOSO) analysis. Removing the ELBO trajectory group drops mean AUC by **0.130**, exceeding the entire gap between XGBoost and SAMA. Predictive entropy is the next most useful group (LOSO drop 0.043). All four attention groups contribute less than 0.006 each. Used alone, the ELBO trajectory achieves solo AUC between 0.67 and 0.84 across domains, approaching SAMA's performance without any subset enumeration.

The shadow conditions reinforce this: ELBO+H (8 dims) reaches 0.843 AUC and 19.2% TPR@1%FPR, while attention-only collapses to 0.525 AUC and 1.6% TPR. Full per-domain tables are in Appendix D.

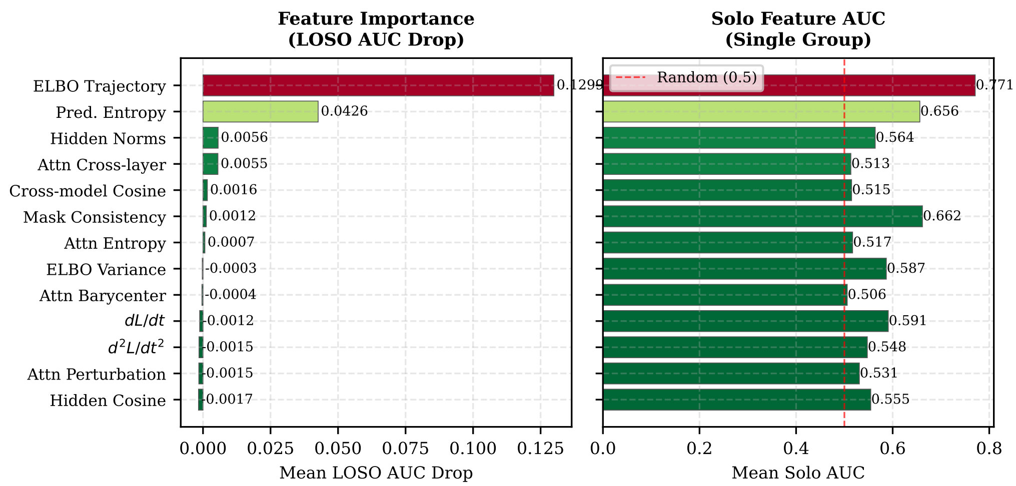

Figure 6: Left: mean LOSO AUC drop per feature group across six domains. Right: mean solo AUC per group using only that group. The ELBO trajectory dominates; all attention groups are near-zero.

6 Conclusion

Fine-tuned MDLMs are significantly more vulnerable to membership inference than grey-box baselines suggest. Extracting a 46-dimensional diffusion-trajectory feature vector at four masking levels and training gradient-boosted classifiers on top achieves mean AUC 0.878 across six MIMIR domains, +0.062 over SAMA. The ELBO trajectory drives most of this: removing it alone drops AUC by 0.130, while all attention features together contribute less than 0.006.

The shadow-model attack makes this threat practical. With $K = 3$ surrogate MDLMs trained on data from different domains, we reach 0.858 mean AUC with no target-domain labels at all. The oracle-to-shadow gap of 0.020 is small enough to consider shadow transfer a viable standalone attack, not just an approximation. Reducing to 8 ELBO+entropy dimensions (Condition C) costs only 0.015 more AUC and makes extraction roughly 6× faster, offering a good trade-off for resource-constrained auditing.

Limitations. All experiments use a single model family (Qwen3-0.6B MDLM) in a fine-tuned setting with small training sets (300–1,000 samples per domain). Behaviour on larger models or in the pretraining regime may differ. The white-box threat model is also optimistic; deployments that restrict internal access would limit the attack to the shadow-model path.

SAMA is evaluated with $T = 4$ progressive masking steps rather than the $T = 16$ used in the original paper, due to compute constraints on the L4 GPU. Since SAMA's score resolution at low FPR thresholds improves monotonically with T [Chen et al., 2026b], the reported $+0.062$ AUC gap and the complete breakdown at 0.1% FPR may partially reflect this underspecification rather than an inherent limitation of the grey-box approach.

Future directions. On the defence side, differential privacy during fine-tuning [Abadi et al., 2016] and explicit regularisation of the ELBO trajectory are natural candidates. On the attack side, extending to LLaDA and other MDLM variants, and estimating the ELBO trajectory from generation-time outputs for fully black-box settings, are open problems.

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A Proof-of-Concept: Run 1 (Single-Domain Overfitting)

Before running the full six-domain pipeline we validated the attack framework on a single controlled experiment: intentionally overfit the MDLM on a small dataset and verify that white-box trajectory features can recover membership under extreme memorisation.

Setup

We fine-tuned `Qwen3-0.6B-diffusion-mdlm-v0.1` on 300 GitHub member texts for **15 epochs** (AdamW, $\eta = 10^{-4}$, batch 8, bfloat16). The intent was to push memorisation hard and stress-test the feature extraction pipeline before scaling to six domains.

Memorisation Verification

The mean member ELBO gap reached **4.625 nats**: the fine-tuned model reconstructs training texts at almost no reconstruction cost at any masking level. The non-member gap was -0.130 nats (base and fine-tuned model are essentially equivalent on unseen text). Figure 7 shows the distribution.

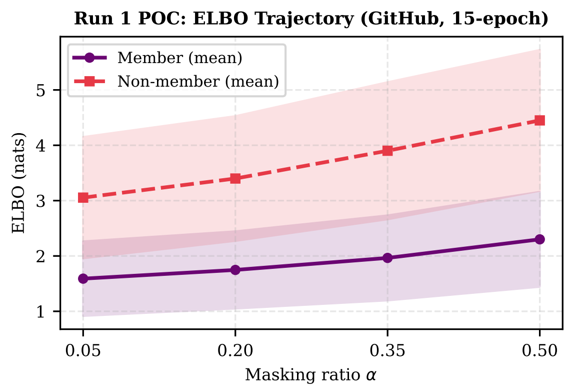

Figure 7: ELBO gap distribution from Run 1 (GitHub, 15 epochs). Member gap mean: 4.625 nats; non-member gap mean: −0.130 nats. The two distributions are nearly fully separated.

Attack Results

Method	AUC	TPR@1%FPR
XGBoost (ours)	**0.989**	**88.9%**
MLP (ours)	0.986	77.3%

Discussion

White-box trajectory features cleanly recover membership even under extreme overfitting. The ELBO curve is consistently lower and more concave for member texts regardless of how low the absolute reconstruction values get, so the classifier can separate the two populations even when the member gap is nearly 5 nats above zero.

This experiment confirmed that the feature extraction pipeline is sound. It also showed that 15 epochs is too many for the main experiments: the model over-memorises in a way that collapses the discrimination into a trivial task and moves the operating point far outside the regime of practical deployments. The main experiments use 5 epochs (see Appendix B), which produces member ELBO gaps of 2.19–2.75 nats across domains and keeps the attack in a more realistic range.

B Hyperparameter Search: Run 2

Run 2 extended the proof-of-concept by introducing all five attack methods (Loss, Zlib, Ratio, SAMA, XGBoost, MLP) on the GitHub domain and varying the number of fine-tuning epochs to find the operating point with the best membership discriminability.

Epoch Ablation

The central finding is that discriminability is *non-monotone* in training epochs. With too few epochs the model has not memorised sufficiently to create an ELBO gap; with too many epochs (15 as in Run 1), over-memorisation causes the SAMA baseline to collapse and makes the domain generally "easy" regardless of membership status. The 5-epoch operating point produced ELBO member gaps of 2.19–2.75 nats across domains—within SAMA's designed operating range—while maintaining strong XGBoost discriminability.

Final Hyperparameter Configuration

Parameter	Value
Fine-tuning epochs	5
Learning rate	10^{-4}
Weight decay	0.01
Batch size	8
Precision	bfloat16
Masking ratios $\boldsymbol{\alpha}$	{0.05, 0.20, 0.35, 0.50}
Masks per ratio K	8
Feature dimensionality	46
SAMA subsets N	128
SAMA steps T	4
XGBoost trees	200
XGBoost max depth	4
XGBoost learning rate	0.05
MLP hidden layers	(256, 128, 64)
MLP dropout	0.3
CV folds	5
Bootstrap samples	1000

All experiments in Run 3 and Run 4 use these exact hyperparameters.

C Full Feature Taxonomy (46 Dimensions)

Table 3 lists all 46 features in the order they appear in the feature vector $\phi(x)$. Features are extracted at each of four masking ratios $\alpha_j \in \{0.05, 0.20, 0.35, 0.50\}$ unless otherwise noted (marked "global"). The *Expected direction* column indicates the sign of the difference $\phi_{\text{member}} - \phi_{\text{non-member}}$ predicted by the memorisation hypothesis; this is used as a sanity check but not enforced during training.

Table 3: Complete 46-dimensional feature vector description. $K = 8$ independent random masks are averaged per (feature, α_j) pair.

Idx	Name	Group	Description	Direction
1–4	`elbo_t{0-3}_a{α}`	ELBO Traj.	Masked NLL $\mathcal{L}(x; \alpha_j)$ at each ratio	member < non-member
5	`elbo_var`	ELBO Variance	Variance of ELBO across K masks at $\alpha_0 = 0.05$	member < non-member
6–9	`dldt_t{0-3}`	dL/dt	Finite-difference $\partial\mathcal{L}/\partial\alpha$ at each ratio	member < non-member
10–13	`d2ldt2_t{0-3}`	d^2L/dt^2	Second derivative estimate at each ratio	member > non-member
14–17	`pred_entropy_t{0-3}`	Pred. Entropy	Mean $H[\, p_\theta(\cdot \mid x_{\alpha_j})\,]$ at masked positions	member < non-member
18–21	`mask_consistency_t{0-3}`	Mask Consist.	Mean agreement of predictions under two independent masks	member > non-member
22–25	`hidden_norms_t{0-3}`	Hidden Norms	Mean $\lVert \mathbf{h}^\ell \rVert_2$ of final layer hidden state	member < non-member
26–29	`hidden_cosine_t{0-3}`	Hidden Cosine	Cosine sim. between FT and base hidden states at last layer	member > non-member
30–33	`attn_entropy_t{0-3}`	Attn Entropy	Mean entropy of attention weight distribution over heads	member < non-member
34–37	`attn_crosslayer_t{0-3}`	Attn Cross-layer	Pearson correlation of attention maps between consecutive layers	member > non-member
38–41	`attn_barycenter_t{0-3}`	Attn Barycenter	Wasserstein barycenter position of attention mass	domain-dependent
42–45	`attn_perturbation_t{0-3}`	Attn Perturb.	Change in attention entropy under small input token perturbation	member < non-member
46	`cross_model_cos`	Cross-model Cos.	Cosine sim. of last-layer representation: FT vs. base (global)	member > non-member

Table 4: Leave-One-Signal-Out (LOSO) AUC drop per feature group across six domains. Positive values indicate that removing the group hurts performance. **Bold**: largest drop (most important). Full AUC (all features) is 0.877–0.930 depending on domain (Table 1).

Feature Group	**Dims**	arXiv	GitHub	H-News	Pile-CC	PubMed	Wiki	**Mean**
ELBO Trajectory	4	**0.146**	**0.045**	**0.141**	**0.135**	**0.172**	**0.141**	**0.130**
Pred. Entropy	4	0.042	0.014	0.041	0.035	0.064	0.060	0.043
Hidden Norms	4	0.003	0.016	0.001	0.002	0.002	0.010	0.006
Attn Cross-layer	4	–	0.013	0.005	0.009	0.003	0.003	0.006
Cross-model Cos	1	0.001	0.001	–	–	–	0.011	0.002
Mask Consistency	4	–	0.011	0.001	–	–	–	0.001
Attn Entropy	4	–	0.004	0.003	–	–	0.001	0.001
ELBO Variance	1	–	0.001	–	–	–	–	≈0
Attn Barycenter	4	0.001	0.003	–	–	–	–	≈0
dL/dt	4	–	–	–	–	–	0.002	≈0
d^2L/dt^2	4	–	–	–	–	–	–	≈0
Attn Perturbation	4	–	–	–	–	–	–	≈0
Hidden Cosine	4	–	–	–	–	–	–	≈0
Total	*46*							

"–" denotes negligible or negative drop (< 0.001), indicating the feature group is redundant or slightly harmful.

Notes. Features in the ELBO Variance, dL/dt, and d^2L/dt^2 groups show near-zero LOSO drops (Table 4) despite representing distinct mathematical quantities. This is because at $T = 4$ masking levels, the finite-difference estimates of derivatives are nearly collinear with the raw ELBO trajectory values at those four points; removing either set leaves the other to compensate almost perfectly. Extending to $T \geq 8$ levels may break this collinearity and reveal independent signal in the derivative features.

D Full Ablation Study (Run 3)

D.1 LOSO AUC Drop: All Domains

Table 4 shows mean LOSO drops. Below are the full per-domain values.

Table 5: LOSO AUC drop per feature group and domain. Full-model AUC baseline: arXiv 0.877, GitHub 0.801, HackerNews 0.928, Pile-CC 0.930, PubMed 0.842, Wikipedia 0.892.

Feature Group	arXiv	GitHub	H-News	Pile-CC	PubMed	Wiki	Mean
ELBO Trajectory	0.146	0.045	0.141	0.135	0.172	0.141	0.130
Pred. Entropy	0.042	0.014	0.041	0.035	0.064	0.060	0.043
Hidden Norms	0.003	0.016	0.001	0.002	0.002	0.010	0.006
Attn Cross-layer	<0	0.013	0.005	0.009	0.003	0.003	0.006
Cross-model Cosine	0.001	0.001	<0	<0	<0	0.011	0.002
Mask Consistency	<0	0.011	0.001	<0	<0	<0	0.001
Attn Entropy	<0	0.004	0.003	<0	<0	0.001	0.001
ELBO Variance	<0	0.001	<0	<0	<0	<0	<0
Attn Barycenter	0.001	0.003	<0	<0	<0	<0	<0
dL/dt	<0	<0	<0	<0	<0	0.002	<0
d^2L/dt^2	<0	<0	<0	<0	<0	<0	<0
Attn Perturbation	<0	<0	<0	<0	<0	<0	<0
Hidden Cosine	<0	<0	<0	<0	<0	<0	<0

<0 denotes negligible or negative drop (group is redundant).

Table 6: Solo AUC per feature group (using only that group's dimensions) per domain. Random baseline is 0.500.

Feature Group	arXiv	GitHub	H-News	Pile-CC	PubMed	Wiki	Mean
ELBO Trajectory	0.774	0.666	0.828	0.839	0.746	0.793	0.774
Mask Consistency	0.680	0.641	0.703	0.670	0.634	0.665	0.666
Pred. Entropy	0.650	0.601	0.695	0.675	0.651	0.660	0.655
dL/dt	0.581	0.563	0.641	0.624	0.586	0.599	0.599
ELBO Variance	0.590	0.559	0.621	0.611	0.563	0.574	0.586
d^2L/dt^2	0.535	0.520	0.584	0.574	0.535	0.542	0.548
Hidden Cosine	0.541	0.527	0.574	0.563	0.539	0.533	0.546
Hidden Norms	0.527	0.528	0.587	0.585	0.555	0.569	0.559
Attn Perturbation	0.525	0.524	0.553	0.542	0.519	0.519	0.530
Cross-model Cos	0.514	0.506	0.531	0.516	0.513	0.520	0.517
Attn Entropy	0.489	0.513	0.529	0.532	0.498	0.499	0.510
Attn Barycenter	0.494	0.505	0.521	0.516	0.497	0.490	0.504
Attn Cross-layer	0.473	0.529	0.503	0.527	0.491	0.467	0.498

D.2 Solo AUC: All Domains

D.3 Discussion

Derivative features show near-zero LOSO drop. dL/dt and d^2L/dt^2 are finite-difference estimates computed from the same four ELBO values as the trajectory group. At $T = 4$ levels, $dL/dt|_{t_j} \approx \mathcal{L}(t_{j+1}) - \mathcal{L}(t_{j-1})$, which is a linear combination of the trajectory values. The two groups are nearly collinear, so removing either one leaves the other to compensate. Extending to $T \geq 8$ masking levels would break this collinearity.

Attention features are domain-specific. Solo AUC for all attention groups is 0.47–0.53 across most domains, near random. In MDLMs the masked denoising objective produces attention patterns that depend heavily on the specific masking configuration and domain vocabulary, making them inconsistent across the $K = 8$ mask realisations. This is why they also fail under shadow-model transfer (Condition D, AUC 0.525): they contain no signal that generalises across domains.

Minimal effective attack. ELBO trajectory and predictive entropy together (8 dimensions) achieve mean AUC 0.843 in shadow transfer. This is cheaper than the full 46-dim vector and still above SAMA.

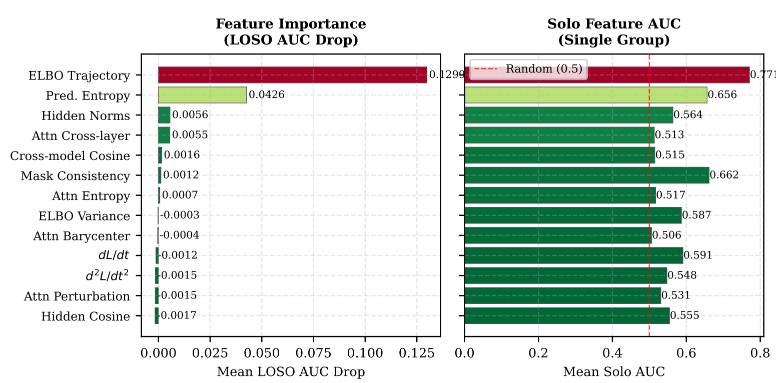

Figure 8: LOSO AUC drop (left) and solo AUC (right), averaged across six domains.

E Shadow Model: Full Results (Run 4)

E.1 Experimental Details

Three shadow MDLMs ($K_s = 3$) are fine-tuned on surrogate data from the `ngram_13_0.2` MIMIR split, which is disjoint from the `ngram_13_0.8` evaluation set. The surrogate pool covers all six domains and contains roughly 5,695 member sequences total, with about 1,000 per shadow model. Non-member texts come from Wikipedia and Common Crawl subsets not used in evaluation. Each shadow model uses the same hyperparameters as the target: 5 epochs, AdamW $\eta = 10^{-4}$, bfloat16.

A single XGBoost classifier is trained on features pooled from all three shadow models and applied to the target domain without any retraining. No target-domain labels are used.

E.2 Full Condition × Domain Results

Table 2 in the main text gives the complete AUC matrix. Figure 9 shows the same data as a heatmap.

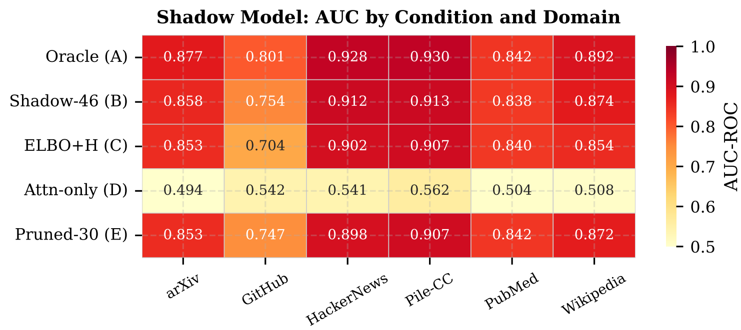

Figure 9: Shadow-model AUC by condition and domain. Condition D (attention only) is near-random universally. Conditions B, C, E all sit above SAMA in most domains.

E.3 Per-Group Transfer AUC

Figure 10 shows the shadow transfer AUC when training on one feature group at a time, analogous to the solo AUC analysis in Appendix D but in the transfer setting.

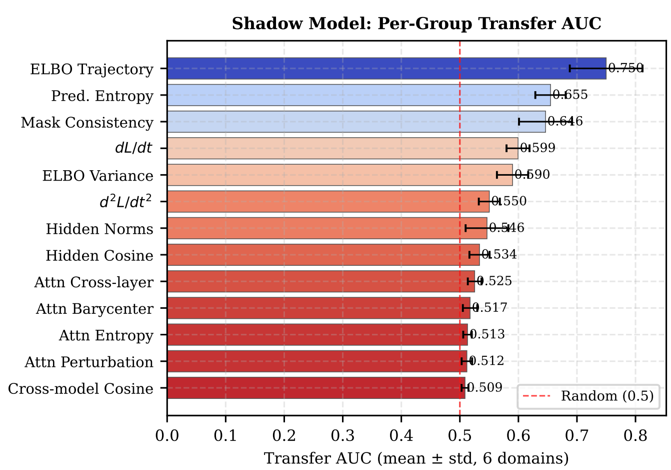

Figure 10: Per-group shadow transfer AUC (mean $\pm$ std across six domains). ELBO trajectory is the only group that transfers above 0.70. All attention groups sit near 0.50.

E.4 Discussion

Why attention features do not transfer. Attention heads in MDLMs specialise to local syntactic patterns of each domain: mathematical notation in arXiv, Python indentation in GitHub, informal pronouns in HackerNews. A classifier trained on these patterns in the shadow domain applies to the wrong pattern space in the target. ELBO trajectory, by contrast, depends mainly on the model's memorisation degree, which is set by the training recipe and generalises across text domains when the architecture is fixed.

ELBO+H as a minimal attack. Condition C (ELBO+H, 8 dimensions) achieves 0.843 mean AUC with roughly $6\times$ less extraction cost than the full 46-dim vector. For practical auditing this offers a strong accuracy-to-cost ratio: 0.843 AUC with no target labels, compared to SAMA's 0.816 AUC which does require black-box model access per sample.

F Per-Domain ROC and PR Curves

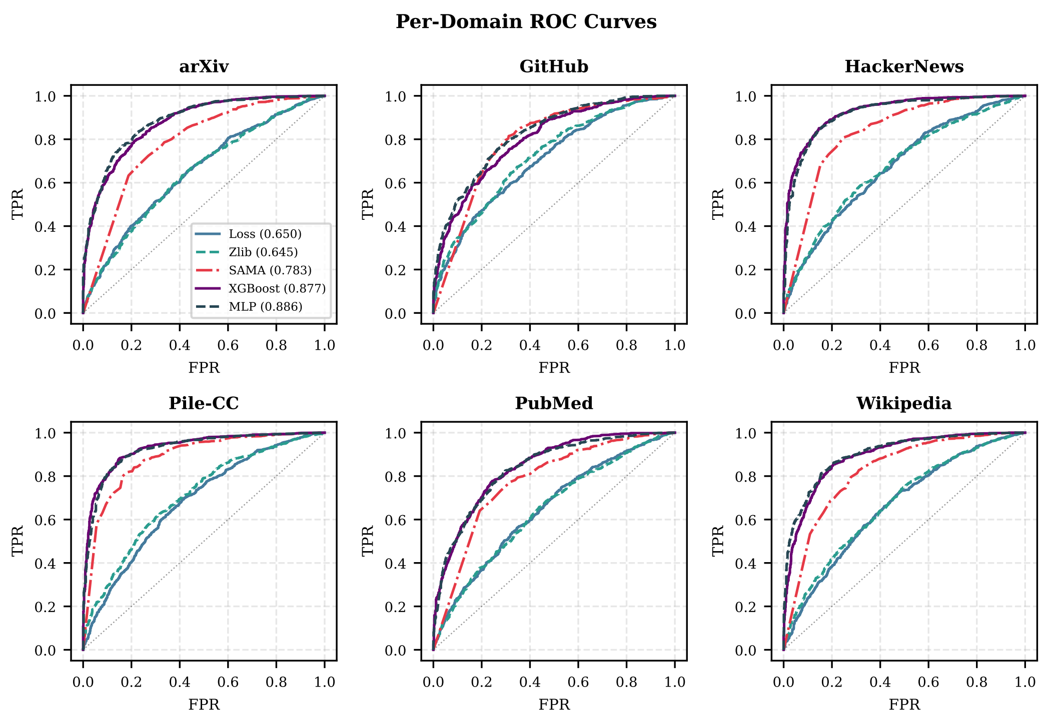

Figure 11: Per-domain ROC curves for all five evaluated methods. The random baseline diagonal is dotted. AUC values appear in the legend of the arXiv panel; method colours are consistent across panels.

Domain notes. **arXiv.** XGBoost and MLP track closely above SAMA at all FPR thresholds.

GitHub. SAMA (AUC 0.795) nearly matches XGBoost (0.801). This is the only domain where SAMA closes to within 0.01 of the white-box classifier, likely because source code has low perplexity and SAMA's NLL comparison works reasonably well.

HackerNews. Largest raw gap: XGBoost 0.928 vs SAMA 0.833. Short, informal sentences with distinctive vocabulary produce particularly clean ELBO trajectories for member texts.

Pile-CC. XGBoost achieves its highest TPR at 10% FPR (0.803) here. SAMA also performs well (0.885) relative to other domains.

PubMed Central. Highly formulaic biomedical text reduces the ELBO curvature difference between members and non-members.

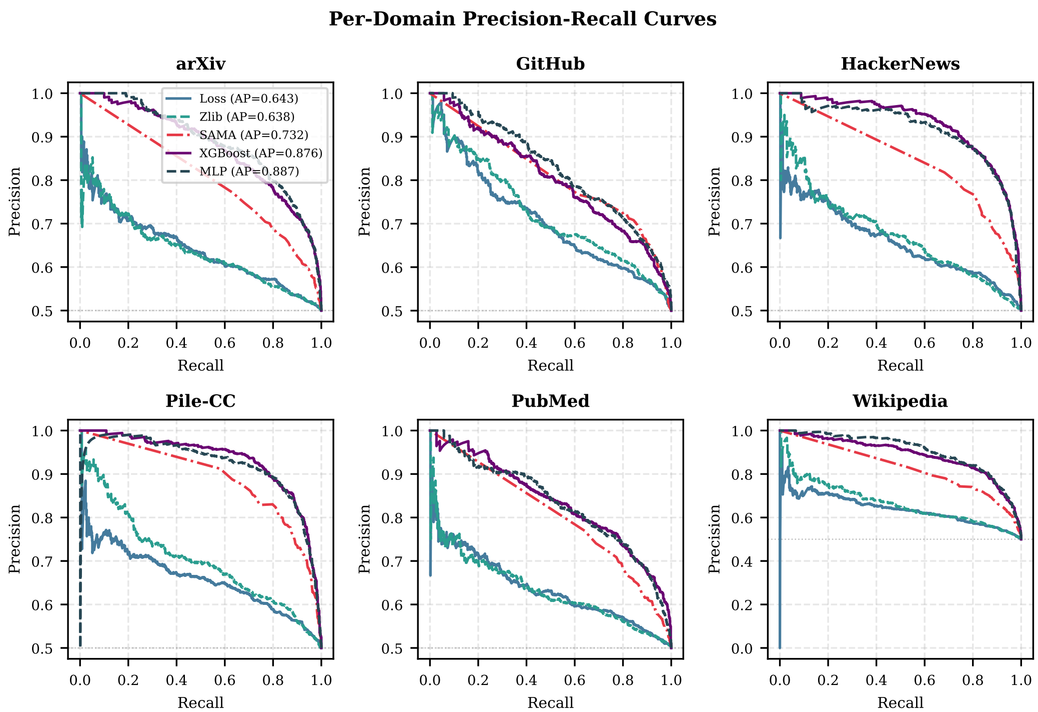

Figure 12: Per-domain precision-recall curves. The horizontal dotted line is the random classifier baseline (class balance = 0.5).

Wikipedia. MLP slightly outperforms XGBoost (0.902 vs 0.892). This is the only domain where MLP takes the top spot; encyclopedic text's consistent structure may favour the MLP's learned feature interactions.

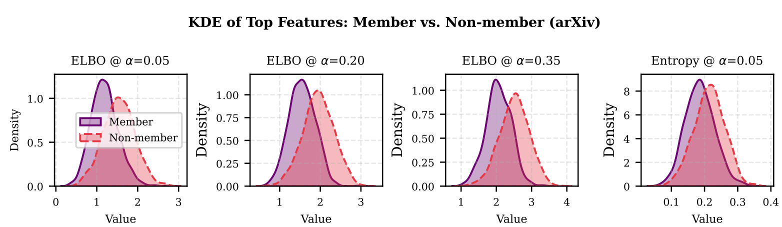

Figure 13: KDE of the four most discriminative features for arXiv. ELBO values at low masking ratios show the clearest separation.